

The reference designs include, for instance, a chipKIT Max32 kit based on Microchip processor and motor control using Fairchild FNB41060.

**ModelSource.** ModelSource is an online library of PCB and schematic components with over 80,000 of them. Designers can add components in their designs from ModelSource library and, while doing so, they can see various parameters of the components, manufacturers' details, data sheets and components preview.

**Simulation interface.** The Simulation interface consists of Simulation Output, SPICE libraries and Simulation Netlist. For Simulation Output, DesignSpark PCB has built-in SPICE parameters that can be assigned to components so that a number of circuit elements could be output to a number of standard SPICE simulators. SPICE simulators supported in DesignSpark PCB are LsSpice, LTspice, B2spice and TINA.

SPICE libraries available contain commonly-used components and symbols, including circuit stimuli and probes to help designers quickly generate circuits to simulate. SPICE Netlists writes a description of the devices in the circuit using basic SPICE format.

Simulation interface lets designers analyse various simulation responses (like AC response, DC response, transient response) of their schematic in one of the above simulators. This saves a lot of time for designers as they need not make a schematic separately in the simulation tool.

**Library creation and Components Wizard.** DesignSpark PCB gives freedom to designers to create new schematic symbols, PCB footprints and join them to form components which can be added to the user-defined library. Also, since schematic symbols and PCB footprints are separate entities in DesignSpark PCB, by using Components Wizard designers can create new components by providing a few parameters to pre-defined schematic symbols and PCB footprint. Components Wizard also helps designers link multiple schematic symbols with different PCB

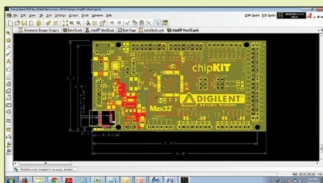


Fig. 2: ChipKIT Max32 reference design

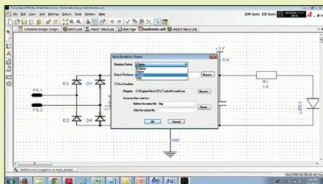


Fig. 3: Spice simulation output

## DesignSpark Mechanical: For free and professional 3D model design

DesignSpark Mechanical is a gesture based modelling tool with an intuitive user interface which lets you create conceptual 3D designs with ease (no need to be a CAD expert). Once you are done with your conceptual 3D designs you can save them in the form of interactive 3D PDF to show it to your customers/clients, or you can even print them using 3D printers.

DesignSpark Mechanical gives a liberty to electronics designers to create conceptual 3D designs of their products themselves in very little time and with very little effort (reducing dependency on external 3D CAD designers). By simply importing their PCB design in DesignSpark Mechanical, designers can create a conceptual 3D design of their product enclosure with very easy-to-use commands and tools. To make it easier, DesignSpark Mechanical has a 3D library of over 38,000 components in various formats, which designers can import directly in their existing designs. There are lots of tutorials, videos and articles available at DesignSpark.com to get you started on DesignSpark Mechanical.

footprints and save the combinations as new components.

**Design calculators.** Track Width calculator lets designers calculate the minimum track width based on the current it carries and temperature rise when current is applied. Track Impedance calculator calculates the characteristic impedance of a unit length of PCB track for a set of common track

geometries. Via Resistance calculator calculates the resistance, voltage drop, power loss and thermal resistance of vias in a PCB design. Other design calculators available are for Heat Sink, RLCC and Conversion.

**Design Rule Check.** While designing a PCB, designers should keep in mind basic points like minimum track width, minimum annular ring, track-track clearance, track-pad clearance and pad-track clearance. Then there are other design rules related to nets, vias, drills and lot many things. When the Design Rule Check is run and errors found, textual error markers get added to the design with the appropriate code for that rule violation. This helps you identify and correct the error.

**Auto-routing.** DesignSpark PCB supports complete auto-routing, partial auto-routing and manual routing. Using an auto-router to route even part of your PCB will give you a significant saving in time. And by combining the power of an auto-router with your experience in PCB layout, the resulting design can beat even the most labour-intensive manual layout results in terms of manufacturability, cost and even aesthetics.